

KẾT QUẢ BÀI THỰC HÀNH 2

Môn học: DS200 - Phân tích dữ liệu lớn

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Họ và tên : Trần Phi Quyên

Bài 1:

```
6 stopwords = LOAD '/ds200-assignments/assignment-2/stopwords.txt' AS (word:c
7
8 -- clean, lowercase, word segment
9 lower_data = FOREACH reviews GENERATE
10 | id,
11 | LOWER(REPLACE(review, '[0-9,.&!%+/\\"?\\\'':()>^~*#=#\\[\\]{}"-]', '')) AS review,
12 | topic,
13 | aspect,
14 | sentiment;
15
16
17 tokenized_data = FOREACH lower_data GENERATE
18 | id,
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 20 bash - assignment-2

tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2\$ pig -x mapreduce Bai1.pig

Output(s):
Successfully stored 177158 records (25005723 bytes) in: "/ds200-assignments/assignment-2/output/bai1_output"

Counters:
Total records written : 177158
Total bytes written : 25005723
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local2110454247_0001

2026-04-10 03:57:00,579 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 03:57:00,582 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 03:57:00,584 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 03:57:00,588 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-10 03:57:00,620 [main] INFO org.apache.pig.Main - Pig script completed in 10 seconds and 470 milliseconds (10470 ms)

tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2\$

Ln 13, Col 12 Spaces: 4 UTF-8

Bài 2:

```
Bai1.pig Bai2.pig assignments.pyynb GenderRatingAnalysis.java MovieRatingAnalysis.java GenreRatingAnalysis.java
[WSL: UBUNTU] ds200-assignments > assignment-2 > Bai2.pig
2
3 raw_data = LOAD '/ds200-assignments/assignment-2/hotel-review.csv' USING PigStorage(';')
4 AS (id:chararray, comment:chararray, topic:chararray, aspect:chararray, category:chararray);
5
6 -- task 1:
7 grouped_words = GROUP clean_words BY word;
8 word_counts = FOREACH grouped_words GENERATE group AS word, COUNT(clean_words) AS count;
9 high_freq_words = FILTER word_counts BY count > 500;
10
11 STORE high_freq_words INTO '/ds200-assignments/assignment-2/output/bai2_word_count' USING PigStorage('\t');
12
13 --task 2:
14 -- sentiment = category
15 grouped_category = GROUP raw_data BY category;

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - assignment-2 + - [ ]

tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ pig -x mapreduce Bai2.pig
Counters:
Total records written : 89
Total bytes written : 24500506
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1431791435_0001
job_local1517222746_0002

2026-04-10 07:18:51,121 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:18:51,123 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:18:51,125 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:18:51,139 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:18:51,143 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:18:51,146 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:18:51,150 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-10 07:18:51,207 [main] INFO org.apache.pig.Main - Pig script completed in 22 seconds and 559 milliseconds (22559 ms)

0 0 0 0 3 20 Java: Activating... Ln 29, Col 24 Spaces: 4 UTF-8 LF [ ] Ph
```

```
17
18 STORE category_counts INTO '/ds200-assignments/assignment-2/output/bai2_category_count' USING PigStorage('\t');
19
20 -- task 3:
21 aspect_data = FOREACH raw_data GENERATE aspect;
22 filtered_aspects = FILTER aspect_data BY aspect IS NOT NULL AND aspect != '';
23
24 grouped_aspects = GROUP filtered_aspects BY aspect;
25 aspect_counts = FOREACH grouped_aspects GENERATE
26   group AS aspect,
27   COUNT(filtered_aspects) AS num_comments;
28
29 STORE aspect_counts INTO '/ds200-assignments/assignment-2/output/bai2_aspect_count' USING PigStorage(';');

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
bash - assignment-2 + - [ ] [ ] [ ] [ ]

tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai2_word_count/part-r-00000
thoai 676
thuận 787
tương 640
điểm 515
lượng 1205

● tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -ls /ds200-assignments/assignment-2/output/bai2_category_count
Found 2 items
-rw-r--r-- 3 tranphiquyen-23521330 supergroup 0 2026-04-10 07:18 /ds200-assignments/assignment-2/output/bai2_category_count/ SUCCESS
-rw-r--r-- 3 tranphiquyen-23521330 supergroup 40 2026-04-10 07:18 /ds200-assignments/assignment-2/output/bai2_category_count/part-r-00000
● tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai2_category_count/part-r-00000
neutral 654
negative 2108
positive 9008

● tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -ls /ds200-assignments/assignment-2/output/bai2_aspect_count
Found 2 items
-rw-r--r-- 3 tranphiquyen-23521330 supergroup 0 2026-04-10 07:18 /ds200-assignments/assignment-2/output/bai2_aspect_count/ SUCCESS
-rw-r--r-- 3 tranphiquyen-23521330 supergroup 99 2026-04-10 07:18 /ds200-assignments/assignment-2/output/bai2_aspect_count/part-r-00000
● tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai2_aspect_count/part-r-00000
HOTEL;3606
ROOMS;2501
SERVICE;2301
```

Bài 3:

```
26 STORE top_positive INTO '/ds200-assignments/assignment-2/output/bai3_top_positive' USING PigStorage(';');
27 STORE top_negative INTO '/ds200-assignments/assignment-2/output/bai3_top_negative' USING PigStorage(';');

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 20 bash - assignment-2

Input(s):
Successfully read 11770 records (15971270 bytes) from: "/ds200-assignments/assignment-2/hotel-review.csv"

Output(s):
Successfully stored 1 records (85748579 bytes) in: "/ds200-assignments/assignment-2/output/bai3_top_positive"
Successfully stored 1 records (85748599 bytes) in: "/ds200-assignments/assignment-2/output/bai3_top_negative"

Counters:
Total records written : 2
Total bytes written : 171497178
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local969652513_0001 -> job_local811101472_0003,job_local1025639699_0002,
job_local811101472_0003 -> job_local1235868844_0004,
job_local1235868844_0004 -> job_local785015590_0006,
job_local785015590_0006 -> job_local1025639699_0002,
job_local1025639699_0002 -> job_local282181199_0005,
job_local282181199_0005 -> job_local1810836211_0007,
```

```
untPair.class 26 STORE top_positive INTO '/ds200-assignments/assignment-2/output/bai3_top_positive' USING PigStorage(';');
27 STORE top_negative INTO '/ds200-assignments/assignment-2/output/bai3_top_negative' USING PigStorage(';');

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 20 bash - assignment-2

2026-04-10 07:43:49,272 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,276 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,277 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,279 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,285 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,287 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,288 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,292 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,295 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,297 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,302 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,303 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,304 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,307 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,308 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,309 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,314 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,316 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,317 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 07:43:49,318 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-10 07:43:49,351 [main] INFO org.apache.pig.Main - Pig script completed in 35 seconds and 685 milliseconds (35685 ms)

tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS 20 bash - assignment-2

tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai3_top_positive/part-r-00000
HOTEL;positive;2942
tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai3_top_negative/part-r-00000
ROOMS;negative;515
tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$
```

Bài 4:

```
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local740648419_0001 ->      job_local55312906_0002,
job_local55312906_0002 ->      job_local1584294568_0003,
job_local1584294568_0003

2026-04-10 09:04:35,466 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,468 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,469 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,473 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,475 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,475 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,479 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,481 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,483 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 09:04:35,486 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-10 09:04:35,521 [main] INFO org.apache.pig.Main - Pig script completed in 30 seconds and 398 milliseconds (30398 ms)

○ tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$
```

```
tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai4_top5_pos_by_category/part-r-000000
PRICES positive cà 1198
PRICES positive vụ 1062
PRICES positive phòng 1007
PRICES positive họp 820
COMFORT positive phòng 1354
COMFORT positive lòng 1004
COMFORT positive khách 989
COMFORT positive hải 985
COMFORT positive sạn 829
GENERAL positive phòng 6989
GENERAL positive khách 5660
GENERAL positive vụ 5475
GENERAL positive tiện 5082
GENERAL positive nhân 4944
QUALITY positive phòng 1041
QUALITY positive chất 803
```

```
tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2$ hdfs dfs -cat /ds200-assignments/assignment-2/output/bai4_top5_neg_by_category/part-r-000000
CLEANLINESS negative không 434
CLEANLINESS negative sạch 243
CLEANLINESS negative sinh 198
CLEANLINESS negative khách 188
MISCELLANEOUS negative không 370
MISCELLANEOUS negative phòng 330
MISCELLANEOUS negative khách 220
MISCELLANEOUS negative nước 162
MISCELLANEOUS negative sạn 147
STYLE&OPTIONS negative sáng 239
STYLE&OPTIONS negative ăn 209
STYLE&OPTIONS negative món 184
STYLE&OPTIONS negative không 182
STYLE&OPTIONS negative bữa 97
DESIGN&FEATURES negative phòng 615
DESIGN&FEATURES negative không 434
DESIGN&FEATURES negative khách 212
DESIGN&FEATURES negative nhỏ 175
DESIGN&FEATURES negative sạn 169
```

Bài 5:

ds200-assignments > assignment-2 >  bai5.pig

```
28 top_neg = LIMIT sorted_neg 5;
29
30 STORE top5_pos INTO '/ds200-assignments/assignment-2/output/bai5_relate_positive' USING PigStorage('\t');
31 STORE top5_neg INTO '/ds200-assignments/assignment-2/output/bai5_relate_negative' USING PigStorage('\t');
```

PROBLEMS OUTPUT DEBUG CONSOLE

TERMINAL

PORTS 

 bash - assignment-2 

```
2026-04-10 08:49:13,385 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,387 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,388 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,388 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,390 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,391 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,392 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,394 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,395 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,396 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,400 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,403 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,404 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,408 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,409 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,411 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,422 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,423 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,424 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-04-10 08:49:13,425 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-04-10 08:49:13,444 [main] INFO org.apache.pig.Main - Pig script completed in 46 seconds and 325 milliseconds (46325 ms)
```

o tranphiquyen-23521330@LAPTOP-7F6ABEPK:~/ds200-assignments/assignment-2\$

Java: Activating

Ln 6, Col 34 Spaces: 4 LITE-8 15